Computer Science & IT



Database Management System

Transactions and Concurrency control

Lecture No. 05



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Recap of Previous Lecture



- Topic: Classification based on recoverability
- Topic: Irrecoverable schedule
- Topic: Recoverable schedule
- Topic: Cascading rollback problem
- Topic: Cascadeless recoverable schedule
- Topic: Strict recoverable schedule

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Topics to be Covered



- Topic: Classification based on serializability
- Topic: Conflicting and non-conflicting operations
- Topic: Conflict equivalent schedule
- Topic: Conflict serializable schedule
- Topic: Precedence graph
- Topic: Topological order

Slide

Note:- In a strict recoverable schedule if transaction Ti updates the dataitem "A", then no other transaction is allowed to read the dataitem A or write the dataitem A, until the Commit/Rollback of transaction Ti

Strict recoverable Schedule

free from

1. Cascading Rollback problem
2. WR problem
3. WW Problem
4. Lost update problem

Not free from

1. RW problem

Note: We are not worried about RW problem or WR problem or WW problem, because this problems can be solved by ensuring that the schedule is a serializable schedule

Note: Cascading rollback problem & lost update problem can be possible in serializable schedules as well

∴ To ensure consistency schedule must be strict recoverable schedule and serializable schedule

it avoids Cascading rollback & lost update

Avoids, RW, WR & WW Problem



Topic : Classification of schedule

Classification of Schedules

Based on Recoverability

1. Irrecoverable Schedule
2. Recoverable Schedule
3. Cascadeless recoverable Schedule
4. Strict recoverable Schedule

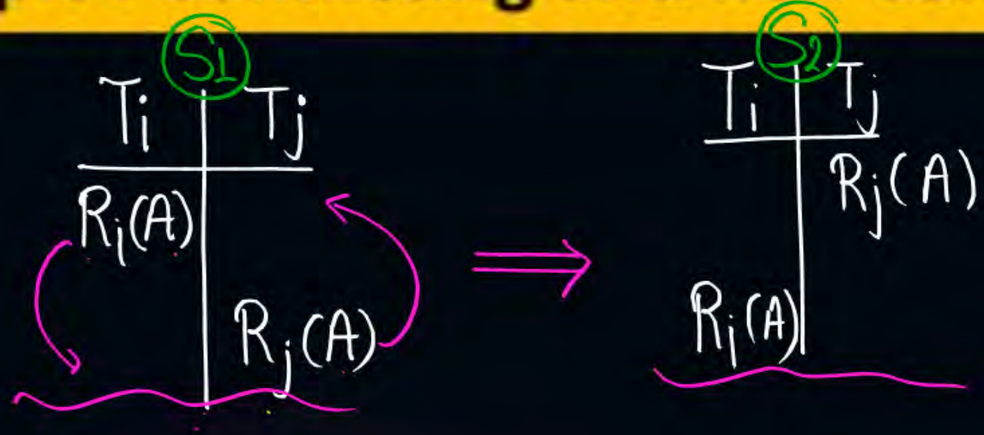
Based on Serializability

1. Conflict Serializable Schedule
2. View Serializable Schedule



Topic : Conflicting and non-conflicting operations

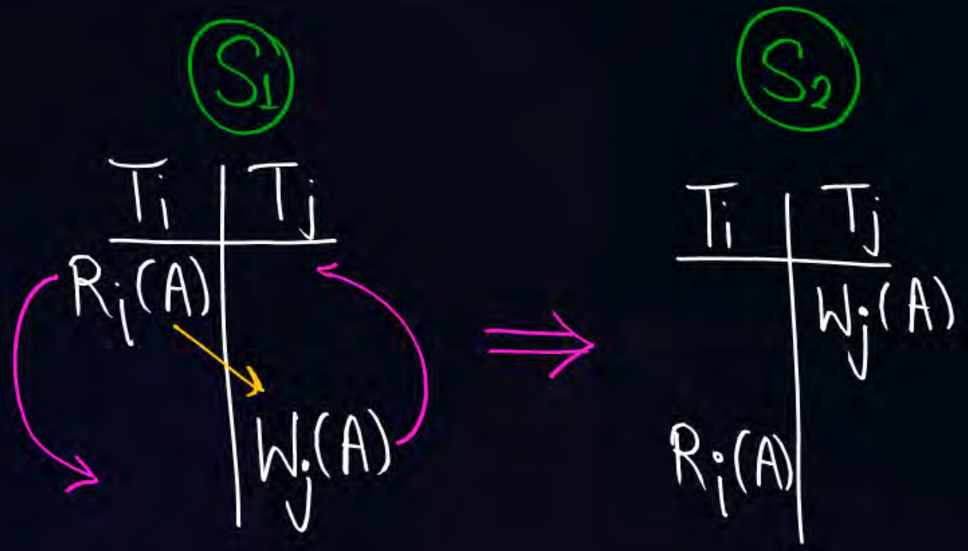
eg:



$S_1 \cong S_2$

∴ $R_i(A)$ and $R_j(A)$ are non-conflicting operations

eg

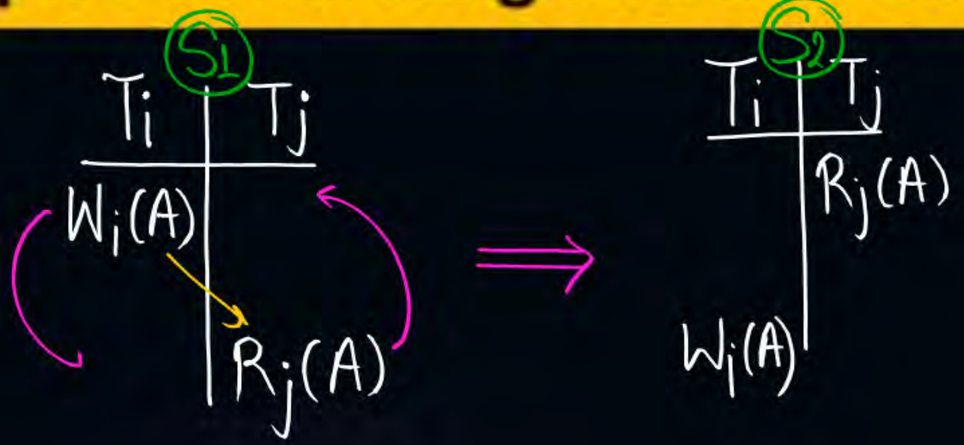


$S_1 \neq S_2$

∴ $R_i(A)$ & $W_j(A)$ are Conflicting operations

Topic : Conflicting and non-conflicting operations

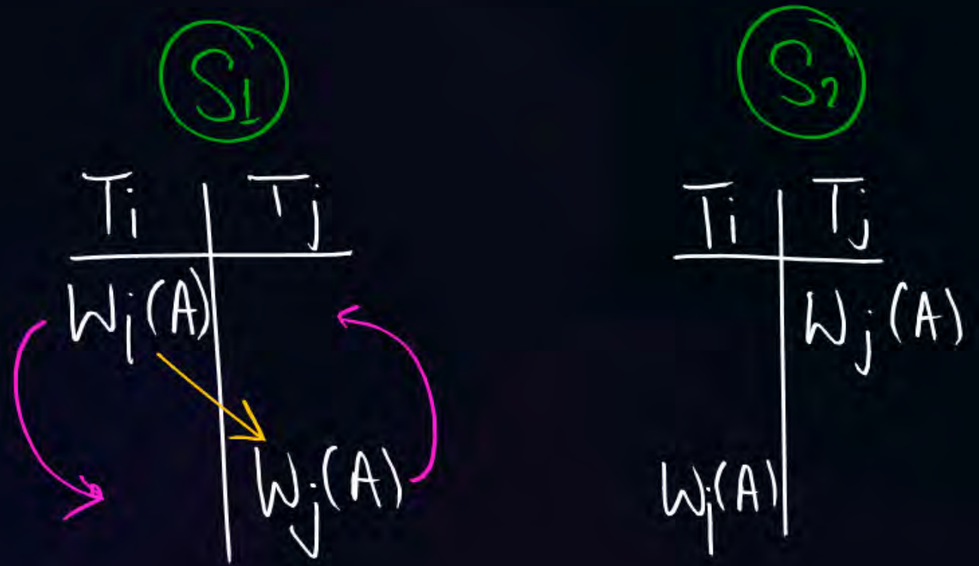
eg:



$S_1 \neq S_2$

∴ $W_i(A)$ & $R_j(A)$ are Conflicting Operations

eg:



$S_1 \not\equiv S_2$

∴ $W_i(A)$ & $W_j(A)$ are Conflicting Operations



Topic : Conflicting and non-conflicting operations

eg:

S_1

T_i	T_j
$R_i(A)/W_i(A)$	
	$R_j(B)/W_j(B)$
	$R_i(A)/W_i(A)$

S_2

T_i	T_j
	$R_j(B)/W_j(B)$

$S_1 \equiv S_2$

∴ $R(A)/W(A)$ are non-conflicting operations w.r.t $R(B)/W(B)$



Topic : Conflicting and non-conflicting operations

Conflicting Operations : Two operations are called as conflicting operations if all the following conditions hold true for them -

- Both the operations belong to different transactions
- Both the operations are on the same data item
- At least one of the two operations is a write operation



Topic : Conflicting and non-conflicting operations

Non-Conflicting Operations : Two operations are non-conflicting operations only if

- Both the operations are on different data items

or

- Both the operations are read operations

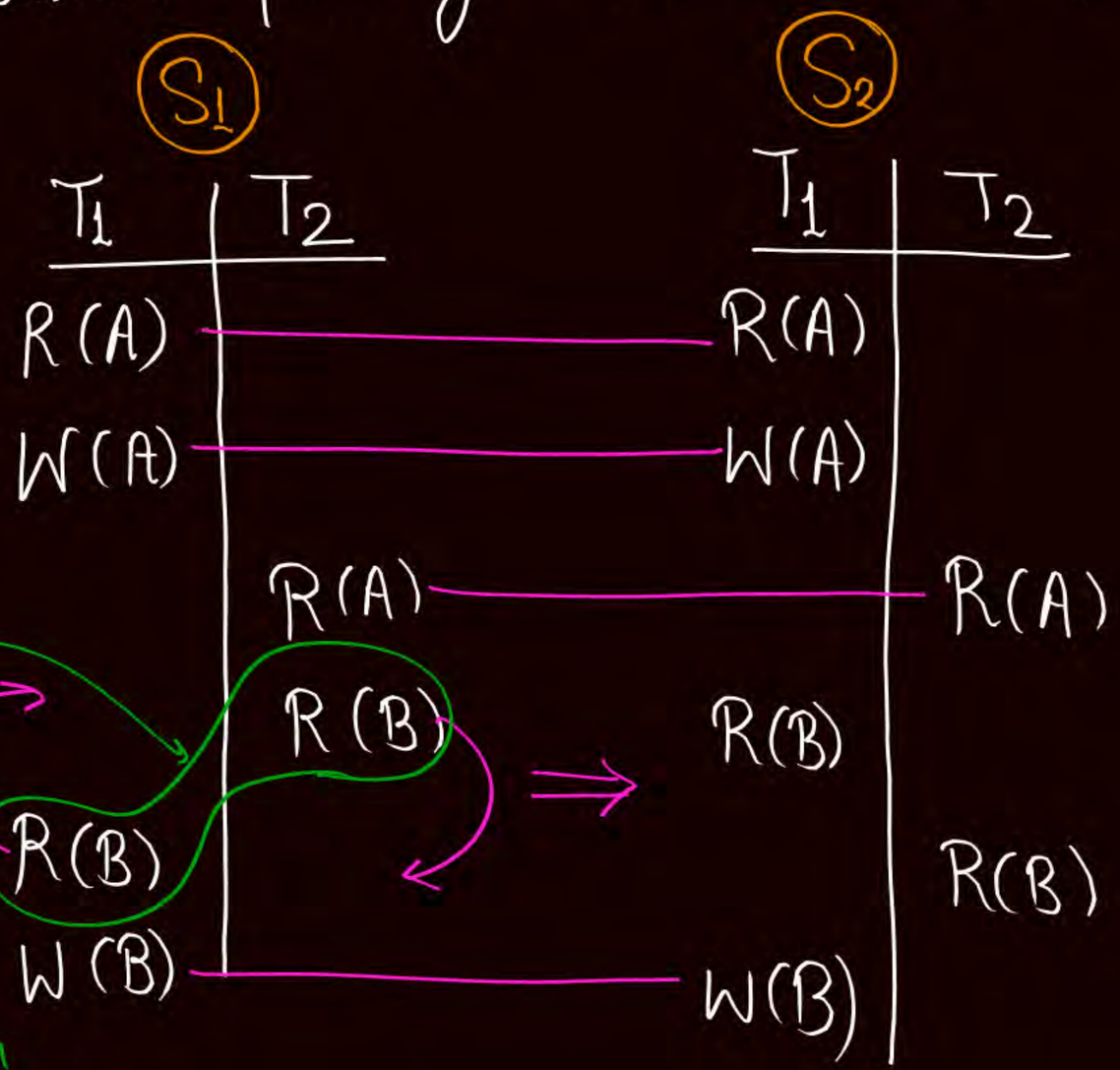


Topic : Conflict equivalent schedule

Consider two schedules Si and Sj,

If one of them can be transformed into another by swapping the position of some of the non-conflicting operations in it then schedule Si and Sj are called conflict equivalent schedule.

eg: Consider following schedules S_1 & S_2



Non-conflicting operations

∴ Their position Can be swapped

S_1 Can be Converted into S_2, by swapping the position of some of the non-conflicting operation of schedule S_1

∴ S_1 is Conflict equivalent to S_2



Topic : Conflict serializable schedule

If a given schedule can be converted into a serial schedule by swapping the positions of some of its non-conflicting operations, then it is called a conflict serializable schedule.

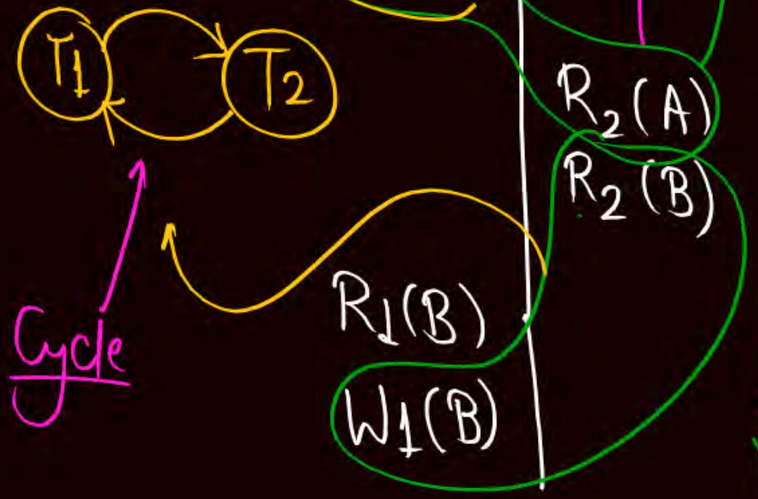
Note: If given schedule is Conflict equivalent to at least one of the serial schedule then, given schedule is a Conflict serializable schedule

Q: Check whether the given schedule is a Conflict serializable schedule or not

S_1 is neither Conflict equivalent to $T_1 \rightarrow T_2$ nor Conflict equivalent to $T_2 \rightarrow T_1$
i.e. S_1 is not Conflict equivalent to any of the serial schedule
∴ S_1 is not a Conflict serializable schedule

S_1

T_1	T_2
$R_1(A)$	
$W_1(A)$	
	$R_2(A)$
	$R_2(B)$
$R_1(B)$	
$W_1(B)$	



$R_2(A)$ is Conflicting operation with $W_1(A)$
∴ $R_2(A)$ can not be moved above the $W_1(A)$
Hence S_1 can not be Conflict equivalent to $T_2 \rightarrow T_1$

$R_2(B)$ is a Conflicting operation with $W_1(B)$ ∴ their respective position can not be swapped i.e, $R_2(B)$ can not be moved below $W_1(B)$. Hence S_1 can not be Conflict equivalent to $T_1 \rightarrow T_2$

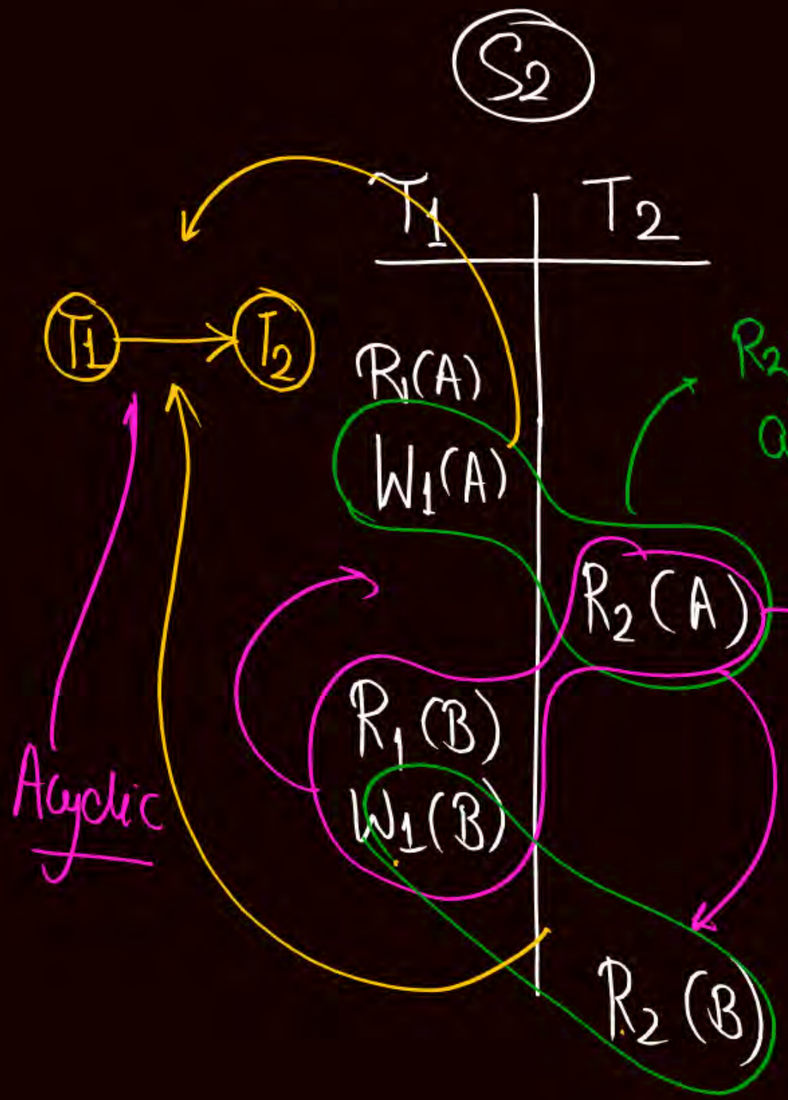
$T_1 \rightarrow T_2$

T_1	T_2
$R_1(A)$	
$W_1(A)$	
$R_1(B)$	
$W_1(B)$	
	$R_2(A)$
	$R_2(B)$

$T_2 \rightarrow T_1$

T_1	T_2
	$R_2(A)$
	$R_2(B)$
$R_1(A)$	
$W_1(A)$	
$R_1(B)$	
$W_1(B)$	

Q: Check whether the given schedule is a Conflict serializable schedule or not



$R_2(A)$ can not be moved above $W_1(A)$

$\therefore S_2$ is not Conflict equivalent to $T_2 \rightarrow T_1$

$R_2(A)$ is non-conflicting with $R_1(B)$ as well as $W_1(B)$

$\therefore R_2(A)$ can be shifted below $R_1(B)$ & $W_1(B)$

$\therefore S_2$ is Conflict equivalent to serial schedule $T_1 \rightarrow T_2$

Hence S_2 is a Conflict serializable schedule and Conflict equivalent serial schedule is $T_1 \rightarrow T_2$

$T_1 \rightarrow T_2$

T_1	T_2
$R_1(A)$	
$W_1(A)$	
$R_1(B)$	
$W_1(B)$	
	$R_2(A)$
	$R_2(B)$

$T_2 \rightarrow T_1$

T_1	T_2
	$R_2(A)$
	$R_2(B)$
$R_1(A)$	
$W_1(A)$	
$R_1(B)$	
$W_1(B)$	

Note: Conflicting operations of the transactions will provide the precedence order in which transactions must be executed.

eg:

T_i	T_j
$R_i(A)$	
	$W_j(A)$

} In this case the precedence order is T_i then T_j

Otherwise T_i will start reading the value of dataitem A updated by T_j, which will not be same as given schedule



Topic : NOTE

Note 1: If schedule is a Conflict Serializable schedule, then that schedule is a Serializable schedule

IMP Note 2: If schedule is not a Conflict Serializable schedule then it may or may not be Serializable.



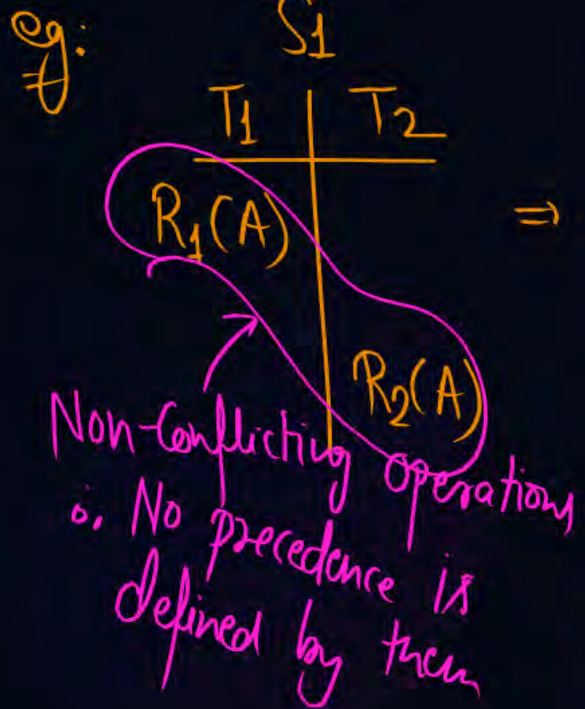
Topic : Precedence graph

Precedence graph $G = (V, E)$ is a directed graph

Where $V = \{$ Transactions of the schedule $\}$

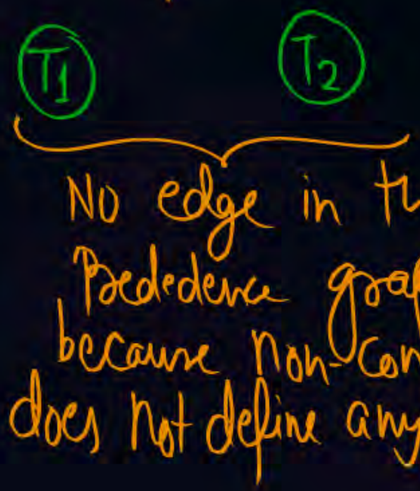
$E = \{$ directional edges based on precedence order specified by Conflicting operations of transactions $\}$

eg:



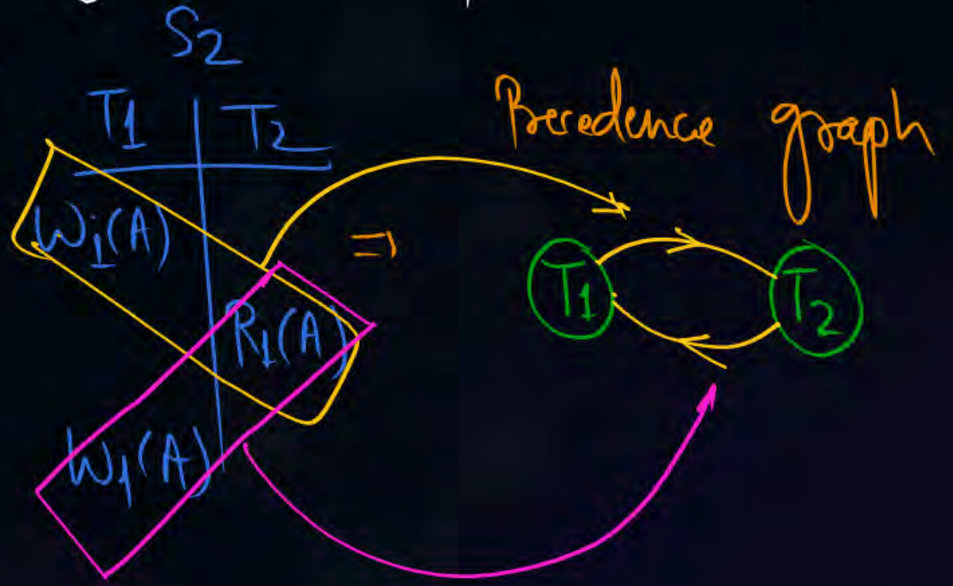
$\Rightarrow$

Precedence graph



No edge in the Pededence graph because non-conflicting oph does not define any Precedence

eg:





Topic : Testing condition for conflict serializable schedule

① If Precedence graph for a Schedule is Cyclic then, that Schedule is not a Conflict serializable Schedule {May or may not be Serializable}

② If Precedence graph for a Schedule is acyclic then, Schedule is a Conflict serializable Schedule, and Conflict equivalent serial Schedules can be given by topological orders of the precedence graph



Topic : Topological order

Let $G_1=(V,E)$ be a directed graph.

Step 1: Visit a vertex $V \in G_1$ such that in-degree of vertex $V=0$ and delete that vertex from the graph.

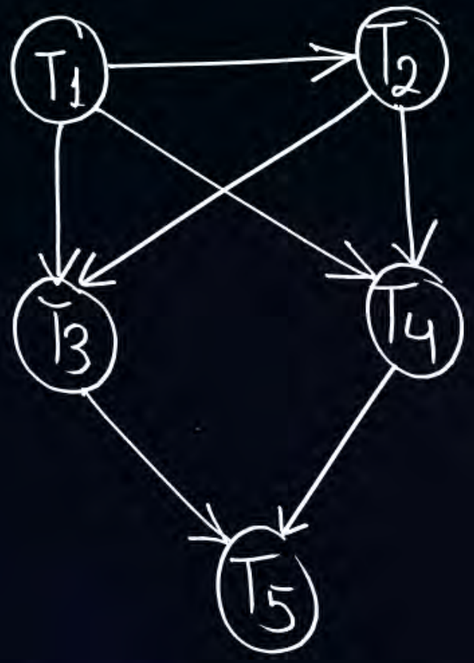
Step 2: Repeat step-1 until graph becomes empty
{i.e. all the vertices are deleted from the graph}

- The order in which the vertices of the graph can be visited is/are topological orders w.r.t. given precedence graph



Topic : Topological order

eg. Consider the following precedence graph



Identify the topological order

$T_1 \longrightarrow T_2 \longrightarrow T_3 \longrightarrow T_4 \longrightarrow T_5$

$T_1 \longrightarrow T_2 \longrightarrow T_4 \longrightarrow T_3 \longrightarrow T_5$

There are two topological orders for the given precedence graph

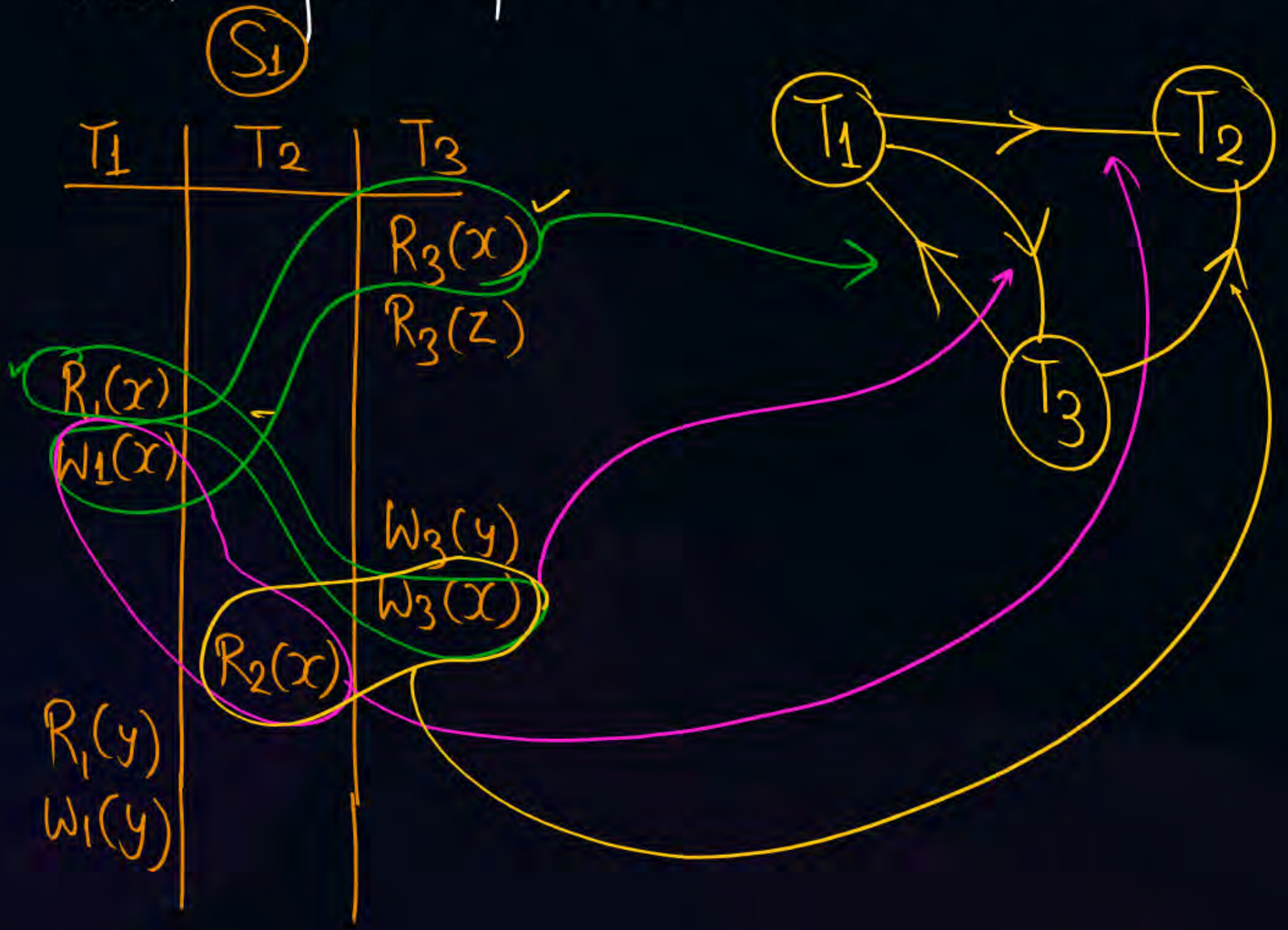
(i) $T_1 \longrightarrow T_2 \longrightarrow T_3 \longrightarrow T_4 \longrightarrow T_5$

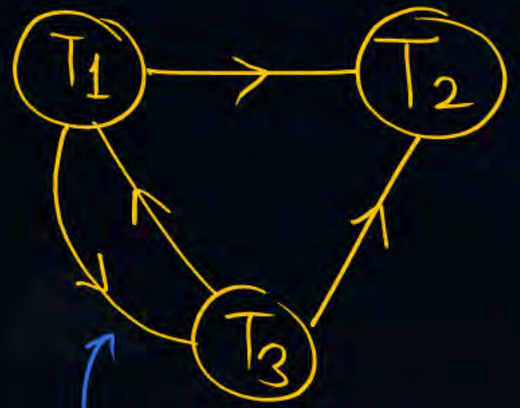
(ii) $T_1 \longrightarrow T_2 \longrightarrow T_4 \longrightarrow T_3 \longrightarrow T_5$

* The schedule corresponding to the given precedence graph is conflict equivalent to these two serial schedules

Q: Check whether the given schedule is Conflict Serializable schedule or not?

If schedule is Conflict serializable schedule then identify all Conflict equivalent serial schedules.





Cyclic precedence graph

∴ Schedule is not Conflict Serializable schedule

Q: Check whether the given schedule is Conflict Serializable schedule or not?
If schedule is Conflict Serializable schedule then identify all Conflict equivalent serial schedules.



2 mins Summary

- **Topic** Classification based on serializability
- **Topic** Conflicting and non-conflicting operations
- **Topic** Conflict equivalent schedule
- **Topic** Conflict serializable schedule
- **Topic** Precedence graph
- **Topic** Topological order

THANK - YOU

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